

Faiyaj Rahman

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EDUCATION

University of Michigan

B.S.E. in Computer Engineering

Ann Arbor, MI

May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Computer Organization, Object Oriented Programming, Discrete Math, Computing Systems, Computational Linear Algebra, Vector Calculus

HONORS & AWARDS

Jane Street FOCUS Participant

Competitive quantitative trading and software engineering program, selected from 1000+ applicants

May 2026

New York, NY

WORK EXPERIENCE

Software Engineering Intern

Bosch | Agentic AI, Python, AWS, RESTAPI, CI/CD, Docker, Kubernetes, Bash, Git

May 2026 – Present

Farmington Hills, MI

- Architecting a modular platform hosting all organization file-conversion tools, first shipping CAN .asc to MF4 with DBC channel mapping for MDA; drops the licensed CANalyzer dependency, saving **20+** engineer-hours monthly.
- Prototyping org-wide agentic AI by wiring LLM agents to internal CLIs and Git, cutting development time by **30%**.
- Deployed a fleet monitoring platform to AWS and scripted multi-tenant vehicle analytics scaling to **5000+** vehicles.
- Engineered a multi-threaded GUI used by **100+** developers to build Python executables, saving **15K** dollars costs.

Software Developer

Saf · Darul Uloom Michigan | Claude, TypeScript, React, Supabase, PostgreSQL, Tailwind, Git

Nov. 2025 – Present

Warren, MI

- Built and deployed the beta version of a school-management system, implementing a normalized Postgres schema, Express REST API, and React client that gated role-based access across admin, teacher, parent, and student views.
- Implemented assignment making, grading, attendance, and an admin panel, replacing paper grades for **20+** educators.
- Ran the school's live pilot on the beta for **3** months, shipping **5** iterations on the grading flow from classroom feedback.
- Currently contributing to the production platform and implementing AI feature work ahead of scholar-backed launch.

Software Engineer

MRacing FSAE | Cursor, C/C++, Embedded Systems/Software, Linux, CAN, Raspberry Pi, Gitlab

Aug. 2025 – Present

Ann Arbor, MI

- Built a C++ Raspberry Pi dashboard decoding **1000+** CAN messages/sec into signals and fault states across **3** pages.
- Refactored the legacy Python GUI codebase into C++, cutting latency $\sim 100\times$ and improving signal timing $\sim 50\%$.
- Collaborated with engineers in testing to debug/validate software, ensuring $\sim 100\%$ data consistency across signals.

Undergraduate Research Software Developer

University of Michigan Electrical Vehicle Center | Claude, Python, Raspberry Pi, Figma, Simulink

Sep. 2025 – May 2026

Ann Arbor, MI

- Developed open-source Python software for an EV charging module, contributing to **\$10k** in net savings for an OEM.
- Designed a Simulink FSM for system-safety logic and charging protocols, porting it to Python for pipeline integration.

PROJECTS

Overround | C++, Python, PyTorch, React, D3.js, FastAPI, PostgreSQL

Jul. 2026 – Present

- Building a transformer with learned club embeddings over **18,000+** matches across five domestic leagues, solving cross-league calibration to predict UEFA Champions League outcomes, outperforming Dixon-Coles and Elo baselines.
- Engineering a multithreaded C++ Monte Carlo engine simulating the **36**-team league phase at **5,000+** sims per sec.

Limit Order Book & Matching Engine | Claude, C++, Linux, Git

May 2026 – Present

- Built a high-performance C++ limit order book with price-time priority, handling **10K–100K+** orders/sec for market orders, cancellations, modifications, and partial fills, achieving **sub-millisecond** execution for every core operations.
- Optimized execution using cache-efficient data structures, custom allocators, and tight memory layouts, reducing latency by $\sim 40\text{--}60\%$ and validating throughput against **1M+** simulated order events under live-market conditions.

Low-Level Modulator | Verilog, Assembly, FPGA, Digital Logic

Jan. 2026 – Apr. 2026

- Engineered a **18**-instruction processor in Verilog on a DE2-115 FPGA driving a synthesizer with **40** assembly modules.
- Designed an integer-math DSP pipeline synthesizing **32-bit** audio from **3** wavetables with **5** modulations at **30 FPS**.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, TypeScript, Assembly, Java, SQL, Verilog, OCaml, MATLAB

Tools/Frameworks: Claude, Cursor, Codex, React, PostgreSQL, Linux, Git, Node, STM32, Raspberry Pi, Altium, CAD